Topological Barriers in Geometric Complexity Theory: Milnor-Fiber Padding Collapse and the Exact Cohomology of the 3×3 Permanent

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Status: five proved theorems with symbolic and finite-field verification (`tests/`, `examples/reproduce_all.py`). No claim of resolving VP vs. VNP is made; the paper establishes structural barriers against naive Betti-number, torus-quotient, and bare Bernstein–Sato root obstructions for padded permanent orbit closures, and computes the exact unconditional cohomology of the 3×3 permanent Milnor fiber.

Abstract

In Valiant's algebraic complexity theory, separating VP from VNP reduces to showing that the padded permanent pad(Perm_n, m) = z^{m-n} Perm_n(X) does not lie in the GL_{m^2}(ℂ)-orbit closure of the determinant Det_m(Y). Following the refutation of naive representation-theoretic occurrence obstructions for padded polynomials (Ikenmeyer & Panova, 2017; Bürgisser, Ikenmeyer & Panova, 2019), a natural question is whether topological invariants of the Milnor fibers F_{Det_m} = SL_m(ℂ) and F_{pad(Perm_n, m)}—combined with Hamm–Lefschetz subspace restriction and the (2n−1)-dimensional row/column scaling torus action—can obstruct orbit-closure containment.

We prove five results clarifying the exact scope and limits of this topological route:

1 **Padding fibration and m-independence of Betti numbers (Theorem 1).** For all m > n, projection onto the padding variable exhibits F_{pad(Perm_n, m)} → ℂ* as a locally trivial fiber bundle with fiber F_{Perm_n}. Its Leray–Serre spectral sequence degenerates at E_2, yielding b_k(F_{pad(Perm_n, m)}) = b_k(F_{Perm_n}) + b_{k-1}(F_{Perm_n}), completely independent of m. Consequently, no naive Hamm–Lefschetz cohomology injection H^k(SL_m) ↪ H^k(F_{pad}) can hold in degrees k ≥ n^2 + 1 (for instance, b_{m^2-1}(SL_m) = 1 > 0 = b_{m^2-1}(F_{pad}) for all m > n), because the degenerate padding subspace fails Whitney transversality to the projective closure of SL_m at infinity.

2 **Non-freeness of the scaling torus and Borel equivariant torsion (Theorem 2).** The scaling torus T ≅ (ℂ*)^{2n-1} does not act freely on F_{Perm_n}: the stabilizer has dimension n at the identity and permutation matrices, and dimension 1 on dense matrices X ∈ (ℂ*)^{n×n}. The naive quotient F_{Perm_n}/T is therefore not a principal bundle. However, we prove that for all n ≥ 2 the fixed-point locus F_{Perm_n}^T is empty. By Atiyah–Bott–Segal localization, the Borel equivariant cohomology H*_T(F_{Perm_n}, ℂ) is a pure torsion module over H*(BT, ℂ), forcing the top-degree Betti bound b_{n^2-1}(F_{Perm_n}) ≤ ∑_{k=0}^{n^2-2} b_k(F_{Perm_n}).

3 **Primary decomposition of Sing(Perm_3 = 0) (Theorem 3).** Over ℂ, the Jacobian ideal J(Perm_3) of the 3×3 permanent decomposes into **15 prime ideals of Krull dimension s = 3** (3 two-zero-row planes R_{ij}, 3 two-zero-column planes C_{ij}, and 9 cross-minor quadric cones D_{ij}). Unlike the determinant, generic rank-1 matrices are smooth points of {Perm_3 = 0}. By Kato's connectivity theorem, F_{Perm_3} ⊂ ℂ^9 is **4-connected** (b_1 = b_2 = b_3 = b_4 = 0).

4 **Exact point count, mixed Tate stratification, and Betti numbers of F_{Perm_3} (Theorem 4).** Projecting onto the first two rows exhibits F_{Perm_3} as an affine 𝔸^2-bundle over U = ℂ^6 \ V, where V = {C_{31} = C_{32} = C_{33} = 0} admits a disjoint stratification into linear and toric pieces: ℂ^3 ⊔ (ℂ^3 \ {0}) ⊔ _{l=1}^3 (ℂ*)^2 ⊔ _{k=1}^3 (ℂ*)^3. This proves unconditionally that F_{Perm_3} is mixed Tate, yields the exact finite-field point count |F_{Perm_3}(𝔽_q)| = q^8 − 5q^5 + 6q^4 − 3q^3 + q^2 = |SL_3(𝔽_q)| + q^2(q−1)^4 for all char(𝔽_q) ≠ 2, and—via full-rank stratified boundary maps δ_4, δ_3, δ_2—establishes the **exact Betti spectrum**:

b(F_{Perm_3}) = (1, 0, 0, 0, 0, 5, 6, 3, 1), total Betti number ∑ b_k = 16 (versus 4 for SL_3(ℂ)), and Euler characteristic χ = 0.

1 **Codimension-1 padding singularity and Bernstein–Sato degeneration counterexample (Theorem 5).** For m − n ≥ 2, the factor z^{m-n} creates a codimension-1 singular locus {z = 0}, driving generic slice Milnor numbers to infinity and depressing the log canonical threshold at the origin to lct_0 = 1/(m−n) < 1 = lct_0(Det_m), which is compatible with lower semicontinuity. While disjoint-variable factorization preserves

intrinsic Bernstein–Sato roots via Roots(b_{g·h}) = Roots(b_g) ∪ Roots(b_h), bare Bernstein–Sato roots do not satisfy subset containment under specialization, as shown by the explicit counterexample f_t = x^2 − t y^3 → x^2.

1. Background and problem setting

Let Det_m(Y) denote the determinant of an m×m matrix of indeterminates and Perm_n(X) the permanent of an n×n matrix. In multiplicative padding form, Valiant's VP ≠ VNP conjecture asks whether for m = poly(n),

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pad(Perm_n, m)(X, z) = z^{m-n} Perm_n(X)  ∉  overline{GL_{m^2}(ℂ) · Det_m}.
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The Geometric Complexity Theory (GCT) program (Mulmuley & Sohoni, 2001, 2007) originally proposed separating these orbit closures via representation-theoretic occurrence obstructions. After Ikenmeyer & Panova (2017) and Bürgisser, Ikenmeyer & Panova (2019) proved that occurrence obstructions cannot even surpass the classical determinantal complexity bounds due to the padding factor, attention turned toward multiplicity obstructions and geometric/topological invariants of the hypersurfaces and their Milnor fibers:

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F_{Det_m} = { Y ∈ M_m(ℂ) : Det_m(Y) = 1 } = SL_m(ℂ) ≃ SU(m),
F_{Perm_n} = { X ∈ M_n(ℂ) : Perm_n(X) = 1 },
F_{pad(Perm_n, m)} = { (X, z) ∈ M_n(ℂ) × ℂ : z^{m-n} Perm_n(X) = 1 }.
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The cohomology ring of F_{Det_m} = SL_m(ℂ) is the exterior algebra Λ_ℂ[y_3, y_5, ..., y_{2m-1}], so its Betti numbers b_k(SL_m) = P_{odd}(k, m) count partitions of k into distinct odd integers in {3, 5, ..., 2m−1}, with total Betti number 2^{m-1}. A natural topological separation strategy attempts to bound b_k(F_{Perm_n}) or b_k(F_{pad(Perm_n, m)}) using the (2n−1)-dimensional row/column scaling torus of the permanent and compare it against b_k(SL_m) via Hamm's affine Lefschetz theorem (Hamm, 1983). Sections 2–6 analyze each step of this strategy rigorously.

2. Padding fibration and failure of naive Hamm–Lefschetz restriction

2.1 Locally trivial fiber bundle over ℂ*

Lemma 1. *For all integers m > n ≥ 1, the projection π: F_{pad(Perm_n, m)} → ℂ, (X, z) ↦ z, is a locally trivial algebraic fiber bundle with fiber F_{Perm_n}.*

Proof. On F_{pad(Perm_n, m)}, the equation z^{m-n} Perm_n(X) = 1 forces z ≠ 0. Because Perm_n(X) is homogeneous of degree n, locally on any analytic or étale chart of ℂ* where a branch of z^{(m-n)/n} is chosen, the rescaling (X, z) ↦ (z^{(m-n)/n} X, z) trivializes π with fiber {Perm_n(W) = 1} = F_{Perm_n}. Globally, passing to the n-fold cyclic cover w ↦ w^n = z of ℂ* shows that the monodromy of the bundle is of finite order n, hence acts trivially on rational/complex cohomology because it is isotopic to the identity inside the connected group C* acting by scalar multiplication. ∎

Theorem 1 (Padding Betti formula and Hamm–Lefschetz collapse). *For all m > n and all degrees k ≥ 0 (with b_{-1} = 0), the Betti numbers of the padded permanent Milnor fiber are independent of the padding dimension m and satisfy*

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b_k(F_{pad(Perm_n, m)}) = b_k(F_{Perm_n}) + b_{k-1}(F_{Perm_n}).
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Consequently, for any m > n, there is no injective cohomology restriction H^k(SL_m(ℂ), ℂ) ↪ H^k(F_{pad(Perm_n, m)}, ℂ) in top degree k = m^2 − 1.

Proof. In the Leray–Serre spectral sequence E_2^{p,q} = H^p(ℂ*, R^q π_* ℂ) ⇒ H^{p+q}(F_{pad}, ℂ), the local system R^q π_* ℂ is trivial because the scalar monodromy ∈ μ_n ⊂ ℂ* extends to the connected group ℂ*. Since ℂ* ≃ S^1 has cohomology only in p ∈ {0, 1}, all differentials d_r (r ≥ 2) vanish for degree reasons, so E_2 = E_∞ and H^k(F_{pad}) ≅ H^k(F_{Perm_n}) ⊕ H^{k-1}(F_{Perm_n}).

Since F_{Perm_n} ⊂ ℂ^{n^2} is a smooth affine hypersurface of complex dimension n^2 − 1, Andreotti–Frankel implies b_j(F_{Perm_n}) = 0 for all j > n^2 − 1, so b_k(F_{pad}) = 0 for all k > n^2. By contrast, SL_m(ℂ) ≃ SU(m) has top non-vanishing cohomology b_{m^2-1}(SL_m) = 1. For m > n we have m^2 − 1 > n^2, so an injection H^{m^2-1}(SL_m) ↪ H^{m^2-1}(F_{pad}) would require 1 ≤ 0, a contradiction. The affine Hamm–Lefschetz theorem does not apply to the specific linear subspace L cutting out a padded projection because L is parallel to asymptotic degenerations of SL_m at infinity and violates Whitney transversality on the projective closure. ∎

3. Non-freeness of the scaling torus and Borel equivariant torsion

Let T = {(U, V) ∈ (ℂ*)^n × (ℂ*)^n : ∏_{i=1}^n u_i v_i = 1} ≅ (ℂ*)^{2n-1} act on M_n(ℂ) by (U, V) · X = U X V. Since Perm_n(UXV) = (∏ u_i v_i) Perm_n(X), T preserves F_{Perm_n}.

Theorem 2 (Stabilizer stratification and Borel equivariant torsion). *For all n ≥ 2:*

1 *The action of T on F_{Perm_n} is* **not free***: at the identity matrix I_n (and at every permutation matrix P_σ), the stabilizer Stab_T(I_n) = {(U, U^{-1}) : U ∈ (ℂ)^n} ≅ (ℂ*)^n has dimension n; on the open dense torus (ℂ*)^{n×n} ∩ F_{Perm_n}, the stabilizer {(c I_n, c^{-1} I_n) : c ∈ ℂ*} ≅ ℂ* has dimension 1.*

2 *The fixed-point locus F_{Perm_n}^T is* **empty***: F_{Perm_n}^T = ∅.*

3 *The Borel equivariant cohomology H*_T(F_{Perm_n}, ℂ) is a pure torsion module over R = H*(BT, ℂ) ≅ ℂ[t_1, ..., t_{2n-1}], and the top-degree Betti number satisfies*

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b_{n^2-1}(F_{Perm_n}) ≤ ∑_{k=0}^{n^2-2} b_k(F_{Perm_n}).
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Proof. (1) At X = I_n, U I_n V = I_n ⇔ v_i = u_i^{-1} for all i = 1..n, which automatically satisfies ∏ u_i v_i = 1 and gives a stabilizer isomorphic to (ℂ*)^n. If x_{ij} ≠ 0 for all i, j, then u_i v_j = 1 for all i, j implies u_1 = ... = u_n = c and v_1 = ... = v_n = c^{-1}, a 1-dimensional stabilizer. Thus the action is never free (in fact, the 1-dimensional diagonal subgroup {(c I_n, c^{-1} I_n)} acts trivially on the entire space M_n(ℂ)).

(2) Suppose X ∈ M_n(ℂ) is fixed by all of T. If some entry x_{ij} ≠ 0, then the character χ(U, V) = u_i v_j must be identically 1 on T = ker(Φ), where Φ(U, V) = ∏_{k=1}^n u_k v_k. Hence u_i v_j = Φ(U, V)^a for some a ∈ ℤ. Comparing exponents of u_i gives a = 1, whereas comparing exponents of u_k for k ≠ i (which exists since n ≥ 2) gives a = 0, a contradiction. Thus X = 0; since Perm_n(0) = 0 ≠ 1, F_{Perm_n}^T = ∅.

(3) By the Atiyah–Bott–Segal localization theorem, localizing with respect to S = R \ {0} gives S^{-1} H*_T(F_{Perm_n}, ℂ) ≅ S^{-1} H*_T(F_{Perm_n}^T, ℂ) = 0, so H*_T(F_{Perm_n}, ℂ) is a pure torsion R-module of rank 0. In the Borel Leray–Serre spectral sequence E_2^{p,q} = R^p ⊗ H^q(F_{Perm_n}, ℂ) ⇒ H^{p+q}_T(F_{Perm_n}, ℂ), every row E_2^{*,q} is a free R-module of rank b_q(F_{Perm_n}) that must be killed in rank by differentials d_r (r ≥ 2) before E_∞. Since no differential can enter the top row q = n^2 − 1 from above, the rank b_{n^2-1}(F_{Perm_n}) must be injected by outgoing differentials into rows q ≤ n^2 − 2, yielding the stated inequality. ∎

Matrix size n	dim(T) = 2n−1	dim Stab_T(I_n)	dim Stab_T(dense)	Action free?	F_{Perm_n}^T = ∅
2	3	2	1	False	True
3	5	3	1	False	True
4	7	4	1	False	True
5	9	5	1	False	True

4. Primary decomposition of Sing(Perm_3 = 0)

Theorem 3 (15 prime components of the 3×3 permanent singular locus). *Over ℂ, the singular locus Σ = Sing({Perm_3 = 0}) = V(J(Perm_3)) ⊂ ℂ^9 is the union of* **15 irreducible prime components***, each of Krull dimension s = 3:*

- *3 two-zero-row linear subspaces R_{12}, R_{13}, R_{23} ≅ ℂ^3;*
- *3 two-zero-column linear subspaces C_{12}, C_{13}, C_{23} ≅ ℂ^3;*
- *9 cross-minor quadric cones D_{ij} (1 ≤ i, j ≤ 3), where row i and column j vanish and the complementary 2×2 permanent minor vanishes: D_{ij} ≅ { (y_{11}, y_{12}, y_{21}, y_{22}) ∈ ℂ^4 : y_{11} y_{22} + y_{12} y_{21} = 0 }.*

Consequently, dim_ℂ Sing({Perm_3 = 0}) = 3, and by Kato's connectivity theorem (Kato, 1975) the Milnor fiber F_{Perm_3} ⊂ ℂ^9 is at least (9 − 2 − 3) = **4-connected***, so b_1 = b_2 = b_3 = b_4 = 0.*

Proof. The 9 partial derivatives ∂Perm_3/∂x_{ij} are the complementary 2×2 permanent minors C_{ij}. Each R_{ab} and C_{ab} is defined by 6 linear variables, so ℂ[X]/P ≅ ℂ[t_1, t_2, t_3] is a domain of Krull dimension 3, and every 2×2 minor contains at least one row in {a, b} or column in {a, b}, hence vanishes. Each D_{ij} is defined by 5 linear variables (the 5 entries in row i ∪ col j) together with the non-degenerate rank-4 quadric C_{ij} = y_{11} y_{22} + y_{12} y_{21} = 0, which is irreducible of Krull dimension 4 − 1 = 3. Conversely, if X ∈ Σ has a row (say row 1) and

a column (say col 1) vanishing, C_{11} = 0 places X in D_{11}; if no two rows and no row-column pair vanish, having all 2×2 permanent minors zero in characteristic ≠ 2 forces two rows or two columns to vanish (see Section 5.1). Note that each R_{ab} and C_{ab} lies in the boundary of the cross components D_{ij} where a free coordinate degenerates to 0, which explains why naive affine elimination solvers report 13 branches unless homogeneous primary decomposition is used. Finally, on a generic rank-1 matrix X = u v^T with u_i, v_j ≠ 0, ∂Perm_3/∂x_{11} = 2 u_2 u_3 v_2 v_3 ≠ 0, so unlike Det_3, generic rank-1 matrices are smooth points of {Perm_3 = 0}. ∎

5. Exact point count and unconditional cohomology of F_{Perm_3}

5.1 Affine $\mathbb{A}^2$-bundle projection and toric stratification

Expanding Perm_3(X) along the third row gives

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Perm_3(X) = x_{31} C_{31}(a, b) + x_{32} C_{32}(a, b) + x_{33} C_{33}(a, b) = 1,
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where a = (x_{11}, x_{12}, x_{13}), b = (x_{21}, x_{22}, x_{23}) ∈ ℂ^3 are the first two rows, and

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C_{31} = a_2 b_3 + a_3 b_2,    C_{32} = a_1 b_3 + a_3 b_1,    C_{33} = a_1 b_2 + a_2 b_1.
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Let V = {(a, b) ∈ ℂ^6 : C_{31} = C_{32} = C_{33} = 0} and U = ℂ^6 \ V. For every (a, b) ∈ U, the vector (C_{31}, C_{32}, C_{33}) ∈ ℂ^3 \ {0} is non-zero, so the fiber of π: F_{Perm_3} → U over (a, b) is an affine 2-plane $\mathbb{A}^2$. Hence π: F_{Perm_3} → U is an affine $\mathbb{A}^2$-bundle, and **F_{Perm_3} is homotopy equivalent to U = ℂ^6 \ V**.

Lemma 2 (Disjoint toric stratification of V). *Over any field k of characteristic ≠ 2, V ⊂ k^6 is the union of 5 irreducible 3-dimensional components—two row 3-planes R_1 = {a = 0}, R_2 = {b = 0} and three quadric cones Q_k = {a_k = b_k = 0, C_{3k} = 0} (k = 1, 2, 3)—and decomposes into four pairwise disjoint locally closed strata:*

1. *R_1 ≅ $\mathbb{A}^3$;*
2. *R_2 \ {0} ≅ $\mathbb{A}^3$ \ {0};*
3. *Three 2-tori C_l^∘ = {(a, b) : a_k = b_k = a_m = b_m = 0, a_l ≠ 0, b_l ≠ 0} ≅ (𝔾_m)^2 for {k, l, m} = {1, 2, 3} (where two columns vanish and the remaining column has both entries non-zero);*
4. *Three 3-tori Q_k^∘ = Q_k \ (R_1 ∪ R_2 ∪ C_i ∪ C_j) = {(a, b) : a_k = b_k = 0, a_i, a_j, b_i ∈ 𝔾_m, b_j = −a_j b_i / a_i} ≅ (𝔾_m)^3 for k = 1, 2, 3.*

Proof. Suppose (a, b) ∈ V. If all three columns are non-zero, having any entry zero forces the entire corresponding row to be zero (e.g., a_1 = 0 ⇒ b_1 ≠ 0 ⇒ a_2 = a_3 = 0 via C_{33} = C_{32} = 0), while if all six entries are non-zero, C_{33} = C_{32} = 0 implies b_2/a_2 = −b_1/a_1 = b_3/a_3, and C_{31} = 0 then gives 2 a_2 a_3 (b_2/a_2) = 0, impossible since char(k) ≠ 2. Thus every point of V outside R_1 ∪ R_2 has at least one zero column (a_k, b_k) = (0, 0). If two columns vanish and the third has both entries non-zero, it lies in C_l^∘ ≅ (𝔾_m)^2; if exactly one column (a_k, b_k) vanishes and neither row vanishes, all four remaining entries must be non-zero (else a row or a second column would vanish) and b_j = −a_j b_i / a_i, giving Q_k^∘ ≅ (𝔾_m)^3. ∎

Theorem 4 (Exact point count, mixed Tate structure, and Betti numbers of F_{Perm_3}).

1. *For every finite field $\mathbb{F}_q$ with char($\mathbb{F}_q$) ≠ 2,*

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|V(𝔽_q)| = q^3 + (q^3 − 1) + 3(q − 1)^2 + 3(q − 1)^3 = 5q^3 − 6q^2 + 3q − 1,
|F_{Perm_3}(𝔽_q)| = q^2 (q^6 − |V(𝔽_q)|) = q^8 − 5q^5 + 6q^4 − 3q^3 + q^2 = |SL_3(𝔽_q)| + q^2(q − 1)^4.
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1. *The compactly supported cohomology groups H^m_c(V, ℚ) are pure of a single Tate weight in every degree: H^6_c(V) = ℚ(−3)^5, H^5_c(V) = ℚ(−2)^6, H^4_c(V) = ℚ(−1)^3, H^3_c(V) = ℚ(0)^1, and H^m_c(V) = 0 for m ≤ 2.*
2. *The exact Betti numbers of F_{Perm_3} are*

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b_0 = 1,  b_1 = b_2 = b_3 = b_4 = 0,  b_5 = 5,  b_6 = 6,  b_7 = 3,  b_8 = 1,
Σ_{k=0}^8 b_k(F_{Perm_3}) = 16,      χ(F_{Perm_3}) = 0.
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Proof. Part (1) follows immediately by summing the point counts of the four disjoint strata in Lemma 2. For (2) and (3), let Z = R_1 ∪ R_2 (two ℂ^3 planes meeting at {0}) and W = Z ∪ C_1 ∪ C_2 ∪ C_3 (adding the three column 2-planes C_l ≅ ℂ^2). Since Z, C = ∪ C_l, and Z ∩ C (six coordinate lines ℂ^1 meeting at {0}) are unions of linear subspaces meeting at {0}, the Mayer–Vietoris sequence for W = Z ∪ C yields H^6_c(W) = ℚ(−3)^2, H^5_c(W) = 0, H^4_c(W) = ℚ(−2)^3, H^3_c(W) = ℚ(−1)^6, H^2_c(W) = ℚ(0)^2, and H^1_c(W) = 0.

Next, V \ W = _{k=1}^3 Q_k^∘ ≅ _{k=1}^3 (ℂ*)^3 has H^6_c = ℚ(−3)^3, H^5_c = ℚ(−2)^9, H^4_c = ℚ(−1)^9, H^3_c = ℚ(0)^3. In the long exact sequence of compactly supported cohomology for W ⊂ V, the connecting homomorphisms preserve Hodge weights and reduce to three explicit linear maps (`gct_topology/milnor_permanent.py`):

- δ_4: H^4_c(W) = ℚ(−2)^3 → H^5_c(V \ W) = ℚ(−2)^9 has full column rank 3;
- δ_3: H^3_c(W) = ℚ(−1)^6 → H^4_c(V \ W) = ℚ(−1)^9 has full column rank 6;
- δ_2: H^2_c(W) = ℚ(0)^2 → H^3_c(V \ W) = ℚ(0)^3 has full column rank 2.

Because δ_4, δ_3, δ_2 are all injective, ker(δ_4) = ker(δ_3) = ker(δ_2) = 0, so no mixed-weight extensions or cancellations occur in H^m_c(V): dim H^6_c(V) = 3 + 2 = 5, dim H^5_c(V) = 9 − 3 = 6, dim H^4_c(V) = 9 − 6 = 3, dim H^3_c(V) = 3 − 2 = 1, and H^2_c(V) = 0. Finally, since F_{Perm_3} ≃ U = ℂ^6 \ V is a smooth complex 6-fold homotopy equivalent to U, Poincaré duality on U and the excision sequence for V ⊂ ℂ^6 give b_j(F_{Perm_3}) = dim H^j(U) = dim H^{12-j}_c(U) = dim H^{11-j}_c(V) for all j ≥ 1. ∎

Prime p	Exact |V(𝔽_p)|	Exact |F_{Perm_3}(𝔽_p)|	Exact |SL_3(𝔽_p)|	Difference p^2(p−1)^4	Formula match
3	89	5,760	5,616	144	Exact
5	489	378,400	372,000	6,400	Exact
7	1,441	5,694,192	5,630,688	63,504	Exact
11	5,961	213,637,600	212,427,600	1,210,000	Exact

Degree j	b_j(F_{Perm_3})	b_j(SL_3(ℂ))	b_j(F_{pad(Perm_3, m)})	Cohomological origin in V ⊂ ℂ^6
0	1	1	1	Connectedness of U = ℂ^6 \ V
1	0	0	1	4-connected (dim_ℂ V = 3)
2	0	0	0	4-connected (dim_ℂ V = 3)
3	0	1	0	4-connected (dim_ℂ V = 3)
4	0	0	0	4-connected (dim_ℂ V = 3)
5	5	1	5	H^6_c(V) = ℚ(−3)^5 (5 irreducible 3-folds R_1, R_2, Q_1, Q_2, Q_3)
6	6	0	11	H^5_c(V) = coker(δ_4: ℚ^3 ↪ ℚ^9) = ℚ(−2)^6
7	3	0	9	H^4_c(V) = coker(δ_3: ℚ^6 ↪ ℚ^9) = ℚ(−1)^3
8	1	1	4	H^3_c(V) = coker(δ_2: ℚ^2 ↪ ℚ^3) = ℚ(0)^1
9	0	0	1	Top fiber-bundle shift b_8(F_{Perm_3}) in F_{pad}
Sum	**16**	**4**	**32**	**Euler characteristic χ = 0 in all three cases**

6. Singularity barriers under padding and Bernstein–Sato specialization

Beyond Betti numbers, one may ask whether classical singularity invariants at the origin—such as the log canonical threshold lct_0(f), restricted Milnor numbers μ^{(d)}(f), or the roots of the Bernstein–Sato polynomial b_f(s)—can separate pad(Perm_n, m) from the determinant orbit closure.

Theorem 5 (Padding singularity collapse and Bernstein–Sato root non-monotonicity).

1 *For m − n ≥ 2, the padded polynomial f_{pad} = z^{m-n} Perm_n(X) has singular locus containing the entire hyperplane {z = 0} of codimension 1. Consequently, the restriction of f_{pad} to any generic linear subspace L of dimension d ≥ 2 through the origin has a non-isolated singularity along L ∩ {z = 0}, so the Teissier restricted Milnor number is μ^{(d)}(f_{pad}) = ∞.*

2 *At the origin, lct_0(Det_m) = 1 (since b_{Det_m}(s) = ∏_{k=1}^m (s + k)), whereas by the disjoint-variable product rule,*

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lct_0(z^{m-n} Perm_n(X)) = min(1/(m − n), lct_0(Perm_n)) = 1/(m − n) < 1   (for m − n ≥ 2).
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Because the log canonical threshold is lower semicontinuous under specialization (lct_0(f_0) ≤ lct_0(f_t)), the inequality lct_0(pad(Perm_n, m)) < lct_0(Det_m) is consistent with orbit-closure containment and provides no obstruction.

1 *For any product f(x, y) = g(x) h(y) in disjoint variable sets, the D-module tensor factorization D_{x,y}(g·h)^s ≅ D_x g^s ⊗_{ℂ[s]} D_y h^s implies*

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Roots(b_{z^{m-n} Perm_n}(s)) = { -1, -1/(m-n), ..., -(m-n-1)/(m-n) } ∪ Roots(b_{Perm_n}(s)),
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so the intrinsic Bernstein–Sato roots of Perm_n survive padding unshifted. However, bare Bernstein–Sato roots do **not** *satisfy subset containment Roots(b_{f_0}) ⊂ Roots(b_{f_t}) under 1-parameter degenerations f_t → f_0: for f_t = x^2 − t y^3 → f_0 = x^2, Roots(b_{f_t}) = {−1, −5/6, −7/6} for t ≠ 0, whereas Roots(b_{f_0}) = {−1, −1/2}, and −1/2 ∉ Roots(b_{f_t}).*

Proof. (1) Every partial derivative of z^k Perm_n(X) (k = m−n ≥ 2) contains a factor of z^{k-1}, hence vanishes identically on {z = 0}. Any linear subspace L ⊂ ℂ^{n^2+1} of dimension d ≥ 2 intersects {z = 0} in dimension ≥ d − 1 ≥ 1, so 0 is a non-isolated singularity of f_{pad}|_L. (2) and (3) follow from standard D-module properties of b-functions (Kashiwara, 1976; Saito, 1994) and the explicit b-functions b_{x^2}(s) = (s+1)(s+1/2) and b_{x^2-y^3}(s) = (s+1)(s+5/6)(s+7/6). ∎

7. Conclusion and open problem

The results above close three naive shortcuts in the topological approach to GCT:

- Betti numbers of the padded Milnor fiber F_{pad(Perm_n, m)} decouple completely from the padding size m (Theorem 1), and naive Hamm–Lefschetz injections fail at infinity.
- The row/column scaling torus T^{2n-1} does not act freely on F_{Perm_n} (Theorem 2), though its fixed-point freeness makes H*_T(F_{Perm_n}) a pure torsion H*(BT)-module.
- For n = 3, F_{Perm_3} is 4-connected with exact Betti vector (1, 0, 0, 0, 0, 5, 6, 3, 1) and total Betti number 16 versus 4 for SL_3(ℂ) (Theorems 3 and 4): the permanent Milnor fiber is topologically **richer**, not simpler, than the determinant Milnor fiber, because the 5-component singular locus V ⊂ ℂ^6 introduces 12 extra toric cohomology classes across degrees 5, 6, 7.

Any viable singularity or topological obstruction for VP vs. VNP must therefore work either in the **de-padded slice geometry** Z(Perm_n) = Z(Det_m) ∩ L (requiring universal upper bounds over arbitrary non-generic subspaces L ⊂ M_m(ℂ)) or with semicontinuous invariants of the **Kashiwara–Malgrange V-filtration / Hodge spectrum** on vanishing cycles rather than bare Betti numbers or bare b-function roots.

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